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Promotional Effect of Copper on Activity and Selectivity to Hydrocarbons and Oxygenates for Fischer−Tropsch Synthesis over Potassium-Promoted Iron Catalysts Supported on Activated Carbon

Wenping Ma, Edwin L. Kugler, and Dady B. Dadyburjor*

Department of Chemical Engineering, West Virginia University, Morgantown, West Virginia 26506, United States

ABSTRACT: The effect of 0−2 wt % copper on the activity and selectivity to hydrocarbons and oxygenates was studied over activated-carbon (AC)-supported 15.7 wt % Fe and 0.9 wt % K catalysts for the Fischer−Tropsch synthesis (FTS). The catalysts were examined in a fixed-bed reactor at 300 psig, 3 nL (g of catalyst)$^{-1}$ h^{-1}, and a syngas with a H_2/CO molar feed ratio of 0.9 in a temperature range of 260−290 °C. The addition of up to 2.0 wt % Cu promoter enhances iron reduction significantly. However, the improved iron reduction does not benefit the catalyst activity. Instead, both FTS and water−gas shift (WGS) activities of the Fe−K/AC catalyst are lowered. This may be due to high Cu loadings, suppressing iron carbonization and increasing the deposition of carbon on the catalyst surface during the initial iron carbonization and/or the formation of some Fe−Cu clusters during reduction. The addition of Cu on the Fe−K/AC catalysts changes selectivities to methane and other hydrocarbons only slightly; however, it increases the internal olefin content and decreases the olefin content. This could be attributed to Cu-promoting hydrogen adsorption and enhanced hydrogenation/isomerization. Cu increases oxygenate selectivity of the Fe−K/AC catalysts, implying that Cu enhances molecular CO adsorption, but inhibits CO dissociation on the Fe−K/AC catalyst. The addition of Cu also changes the alcohol distribution of the Fe−K/AC catalysts. The ethanol formation decreases slightly, but the formation of methanol and C_3−C_5 alcohols increases. The highest Cu loading (2 wt %) reduces this trend somewhat. Furthermore, from the kinetic point of view, the addition of 0.8−2.0 wt % Cu decreases both the apparent activation energies and the apparent pre-exponential factors of the FTS and WGS reactions over the catalysts. The decrease in the pre-exponential factor is related to the lower activities of the Cu-containing Fe−K/AC catalysts.

1. INTRODUCTION

Potassium and copper are structural promoters often used for industrial iron-based Fischer−Tropsch synthesis (FTS) catalysts for the formation of heavy hydrocarbons (HCs) and/or oxygenates from synthesis gas (syngas) via the nominal reaction.

$$n\text{CO} + (m + 2n - 2p)/2\text{H}_2 \rightarrow \text{C}_n\text{H}_m\text{O}_p + (n - p)\text{H}_2\text{O} \quad (1a)$$

Extensive studies have been conducted to discover the functions of the promoters in the FTS reaction.[1−10] It is widely accepted that potassium suppresses methane formation and increases the formation rate of heavier HCs through enhanced CO dissociation on iron-based catalysts.[1−3,8] The appropriate amount of potassium, i.e., <1.5 wt %, could significantly increase both FTS activity and the activity for the water−gas shift (WGS) reaction

$$\text{CO} + \text{H}_2\text{O} \leftrightarrow \text{CO}_2 + \text{H}_2 \quad (1b)$$

and could dramatically improve catalyst stability.[1−4] Potassium is also known to increase olefin selectivity greatly and to reduce the secondary reactions of 1-olefins to internal olefins (i.e., olefins where the double bond is not in the terminal position). However, although consistent conclusions about the role of potassium in iron-based catalysts were obtained in many studies, fewer studies deal with the effect of Cu on such catalysts.[5] Furthermore, the results obtained from these studies are scattered, except that Cu decreases the temperature required for the reduction of iron oxides. Bukur et al.,[1] Iglesia and co-workers,[4] and O'Brien and Davis[5] have studied the effects of Cu on precipitated iron catalysts promoted with K, Si, or Zn at 235−270 °C in either a slurry-phase reactor or a fixed-bed reactor. The addition of less than 3 wt % Cu (Cu/Fe atomic ratio less than 0.025) increases both FTS and WGS activities. However, Li and co-workers[2] recently reported that Cu decreases initial WGS and FTS rates when the Cu/Fe atomic ratio is 0.05 (6 wt % Cu). In terms of HC selectivities, Bukur et al.[1] and O'Brien and Davis[5] have reported that Cu decreases methane and increases C_{5+} selectivity; however, Li and co-workers[2] and Iglesia and co-workers[4] have reached the opposite conclusion. O'Brien and Davis[5] and Soled et al.[6] did not see any effect of Cu on olefins and internal olefins; however, Bukur et al.[1] and Li and co-workers[2] observed that Cu suppresses olefin formation and increases internal olefins to some extent. Copper also easily deactivates the catalyst by suppressing CO adsorption and carburization.[2] Furthermore, the activity and stability was reported[1,2] to be improved using both Cu and K promoters.[1,2] The discrepancy of the Cu effects in these studies could be due to different catalyst systems and the different process conditions used.

In a recent study[7] of the role of Cu in the reduction and surface properties of iron-based catalysts, X-ray absorption spectroscopy (XAS) and X-ray photoelectron spectroscopy (XPS) of C(1s) and O(1s) have been used. It was reported that Cu increases Fe dispersion by mixing intimately with Fe_2O_3. At the same time, Cu

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assists in depositing filamentous carbon on the catalyst exposed to H_2 and CO, and this could reduce catalyst activity. Also, the Fe/Cu catalyst could be slightly oxidized under CO/H_2, even though bulk Fe is reduced rapidly. Furthermore, the Cu-containing iron catalyst has an increased XPS signal of O(1s), which reflects an enhanced associative adsorption of CO in the presence of Cu^0, supporting a CO-spillover mechanism, while a large amount of dissociatively adsorbed CO is observed on the Cu-free Fe catalyst. Clearly, the Cu promoter changes the surface chemistry of the iron-based catalysts besides promoting the reduction of iron.

Oxygenates are also important products of FTS. Dependent upon catalyst types and process conditions, the aqueous phase can contain 3−25 wt % oxygenates. However, the effect of Cu on oxygenate formation is little reported to date.

The current effort is aimed at understanding the effects of Cu on iron-based catalysts for FTS, following the same lines as our earlier investigation[3] of the effect of potassium on Fe/AC catalysts. We used a fixed-bed microreactor and a baseline catalyst of 15.7 wt % Fe/0.9 wt % K supported on peat-derived activated carbon (AC) to study how Cu affects catalyst activity, yield, selectivity, and distribution of HCs and oxygenates. Nominal copper loadings of 0−2 wt % are used, and the reaction conditions used are 260−290 °C, 300 psig, 3 nL (g of catalyst)$^{-1}$ h^{-1}, and H_2/CO = 0.9. The experimental observations are explained on the basis of surface species, CO adsorption, and H_2 adsorption.

2. EXPERIMENTAL SECTION

2.1. Catalyst Synthesis. AC from peat (Sigma-Aldrich) was used as a catalyst support. The AC was washed in hot distilled water, calcined at 500 °C for 2 h in flowing N_2, and sieved to 20−40 mesh as the final support for catalyst preparation.

AC-supported potassium-promoted iron catalysts were prepared using incipient-wetness impregnation. Details of preparation procedures can be found elsewhere.[3,9] In brief, an aqueous solution containing ferric nitrate and cupric nitrate corresponding to the required iron and copper contents on the catalyst was first impregnated on the AC support. As mentioned earlier, the nominal loading of Fe was 15.7 wt %. The nominal loadings of Cu used were 0, 0.8, and 2.0 wt %. The resultant materials were dried in air overnight at 90−100 °C. An aqueous solution containing potassium nitrate corresponding to 0.9 wt % potassium was then impregnated onto the prepared Fe/AC or Fe−Cu/AC, again followed by drying in air overnight at 90−100 °C. This produced the three catalysts, nominally 15.7 wt % Fe−0.9 wt % K/AC, 15.7 wt % Fe−0.8 wt % Cu−0.9 wt % K/AC, and 15.7 wt % Fe−2.0 wt % Cu−0.9 wt % K/AC. For convenience of discussion, the abbreviations of 0Cu, 1Cu, and 2Cu represent these three catalysts, respectively.

2.2. Catalyst Characterization. Energy-dispersive spectroscopy (EDS) was used earlier[10] to obtain an elemental analysis of the catalyst materials. The catalysts were also subjected to X-ray diffraction (XRD)[10] and Brunauer−Emmett−Teller (BET) analysis.[18]

In the present work, temperature-programmed reduction (TPR) of the catalysts was conducted in a mixture of 5 wt % H_2−95 wt % Ar at a total flow rate of 0.050 nL/min. About 50 mg of the catalyst was placed in a quartz U-tube (5 × 300 mm) and heated from room temperature (RT) to 915 °C at a constant rate of 10 °C/min. The sample was held at 915 °C for 30 min before quenching. The H_2 consumption rate was monitored by a thermal conductivity detector (TCD).

2.3. Reactor System. The catalyst performance was examined in a computer-controlled down-flow fixed-bed reactor system. A detailed description of the reactor system has been provided previously.[9] Briefly, the flow rate of syngas (H_2/CO = 0.9) containing an internal standard of 5 vol % He was adjusted by a mass-flow controller and directed to a stainless-steel reactor (8 × 630 mm). Typically, about 1.0 g of catalyst (20−40 mesh) was used in the reaction study. The catalyst was diluted 1:4 (v/v) with quartz chips of the same size before being loaded into the reactor.

The catalyst was reduced *in situ* by H_2 at 400 °C, 0.5 MPa, and 3 nL (g of catalyst)$^{-1}$ h^{-1} for 12 h. After the pretreatment, the reactor was cooled to 250 °C in flowing He. The reactor system was then pressurized to 300 psig, the He flow was cut off, and syngas was introduced. Finally, the reactor temperature was gradually brought to 260 °C. After about 20 h time on stream at this temperature, the value was raised to 270 °C and then to 280 °C after an additional 24 h. After another 24 h, catalysts 1Cu and 2Cu were further raised to 290 °C and held there for 24 h.

The exit gas stream (unreacted syngas, CO_2, HCs, water, and alcohols) passed through a product trap (0 °C) to condense liquid products. A liquid sample was generally withdrawn every 24 h. Incondensable gases passed through a back-pressure regulator and then went to either a bubble flow meter or a gas chromatograph (GC) for online product analysis.

2.4. Product Analysis. Inlet and outlet gases were analyzed online every hour by a HP-5890 GC using a HayeSep packed column and a TCD and a DB-1 capillary column and a flame ionization detector (FID), respectively. The organic phase of the liquid sample (containing C_4−C_{34} HCs) was analyzed in a Varian 3400 GC by a MXT-5 capillary column and FID. Branched paraffins were analyzed by a comparison to calibrations using standard samples of C_8 HC. The water phase of the liquid sample was analyzed in the same Varian 3400 GC by a Porapak-Q packed column and FID. The alcohol components in the water phase were identified by matching retention times of known standards. The water-phase composition was quantified using *tert*-amyl alcohol as an internal standard. Mass balance closures were generally 100 ± 3%.

3. RESULTS AND DISCUSSION

3.1. Catalyst Characterization. As mentioned above, the peat-based AC support was subjected to EDS, BET, and XRD in earlier work. In brief, EDS results indicate[10] that trace amounts of Si, Ca, and Mg were detected on the surface, besides oxygen and, of course, carbon as the main element. The BET surface area of the support after N_2 treatment is 606 m^2/g.[18] Of this, 75% is contributed by micropores. The average pore diameter is 5.9 nm.[18] XRD patterns of the spent 1Cu catalyst were obtained[10] in the 2θ range of 30−80°. The phases Fe_3O_4 and Fe_3C_2 were identified, with particle sizes of 28 and 21 nm, respectively.

H_2−TPR patterns of the 0Cu, 1Cu, and 2Cu catalysts determined in the present work are shown in Figure 1. As discussed previously,[3] the TPR pattern of the 0Cu material is characterized by five distinct steps in the temperature range of 25−915 °C. The first three peaks occur in the temperature ranges of 200−320, 320−480, and 480−700 °C, respectively. The first peak is at too high of a temperature to be ascribed to the decomposition of precursor nitrates, which typically occurs at 115−125 °C. Hence, the first two peaks refer to the reductions of iron oxide

$$Fe_2O_3 \rightarrow Fe_3O_4 \rightarrow FeO \qquad (2a)$$

The third peak is almost certainly a reduction to the metal

$$FeO \rightarrow Fe \qquad (2b)$$

For precipitated Fe, the reduction of FeO is rapid and may not be observed in TPR. However, supported Fe is expected to interact with the C support,[12−14] and this is expected to

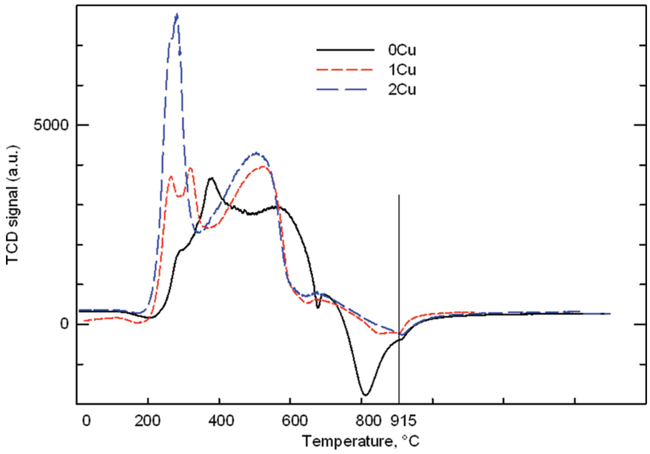

Figure 1. TPR profiles of Fe−K/AC catalysts with and without Cu.

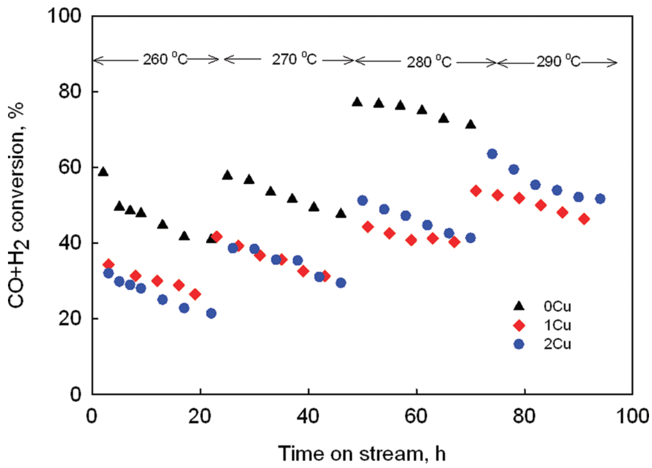

Figure 2. Change of syngas conversion with time on stream over Fe−K/AC catalysts with and without Cu. Other reaction conditions: 300 psig, 3 nL (g of catalyst)$^{-1}$ h^{-1}, and H_2/CO = 0.9.

decrease the rate to the point at which a separate peak is observed.

A smaller peak occurs at about 750 °C. This has been assigned to the reduction of smaller Fe_3O_4 or FeO crystallites located in micropores to metallic Fe, as such crystallites would have a stronger interaction with the AC surface. Finally, a negative peak occurs at about 800 °C. This has been assigned to CO effusion through the reaction.[10]

$$Fe_xO + C \rightarrow Fe_xC + CO \qquad (2c)$$

For the 1Cu catalyst, the reduction temperatures for the first four peaks are shifted down by 20−80 °C, clearly indicating promotion of iron reduction by Cu. However, adding more Cu (2 wt %) to the catalyst results in the first two reduction peaks combining into one very large peak. This suggests that the first reduction step ($Fe_2O_3 \rightarrow Fe_3O_4$ in eq 2a) is too fast to be separately recorded and that the subsequent reduction of Fe_2O_3 to FeO is also fast.

In both the 1Cu and 2Cu catalysts, the third reduction peak is much larger than that of the 0Cu catalyst, further suggesting that Cu increases both the speed and the extent of the reduction of the Fe/AC catalyst. Finally, note that the last peak (negative) of the Cu-containing catalyst in Figure 1 becomes small in absolute value with the addition of Cu. This is probably due to improved iron reduction by Cu, leading to reduced amounts of iron oxides and, hence, smaller amounts of CO emitted.

The significantly improved reduction of iron by Cu has been well-explained by a spillover of H_2 from metallic Cu^0 nuclei to closely associated iron oxide species.[7] This TPR study of Cu-promoted Fe−K/AC catalysts is consistent with results reported in the literature.[2,4,7]

3.2. Catalytic Activity and Stability. The effects of copper loading on catalyst activity and stability are shown in Figure 2 and are averaged over time and temperature ranges in Table 1. Table 1 also shows the values of the rates of FTS (r_{FTS}) and WGS (r_{WGS}) of the Fe−K/AC catalysts with and without Cu, estimated in terms of the rate of formation of CO_2, r_{CO_2}, and the rate of loss of CO, $-r_{CO}$, by

$$r_{FTS} = (-r_{CO}) - r_{CO_2} \qquad (3a)$$

and

$$r_{WGS} = r_{CO_2} \qquad (3b)$$

Turnover frequencies (TOFs) based on the conversion values of Table 1 are shown in Table 2, using values of the metal dispersion and degree of reduction estimated earlier.[10] TOF values in Table 2 are consistent with values in the literature, as also shown earlier.[10]

As expected, all three Fe−K/AC catalysts show increased syngas conversion with increasing temperatures from 260 to 290 °C. At 260 and 280 °C, the 0Cu catalyst shows higher conversions of syngas than the 1Cu catalyst. However, the 2Cu catalyst does not display a marked difference in activity compared to the 1Cu catalyst. As the reaction temperature is increased, the 2Cu catalyst displays a higher syngas conversion than the 1Cu catalyst in the test period of 45−96 h. At all reaction temperatures, deactivation was observed over all three Fe−K/AC catalysts, with or without Cu, but the deactivation rates (reflected by the slopes of the conversion versus time curves of Figure 2) are nearly the same. This clearly suggests that the Cu addition does not alter the stability of the Fe−K/AC catalyst.

The activity of the catalyst decreases with Cu addition, even though Cu significantly increases iron reduction. This could result from one of several possible factors. C(1s) XPS indicates[7] that graphitic carbon and carbonate-like species are observed on a Cu-containing Fe-based catalyst exposed to H_2/CO gas at 275 °C, while these species are not found on a Fe catalyst containing no Cu. This indicates that the Cu promoter fosters deposition of carbon on the catalyst surface, at least at that temperature. Consistent with this explanation, Table 1 indicates higher CH_4 selectivity on the 1Cu and 2Cu catalysts than on the 0Cu catalyst when syngas is initially introduced into the reactor at low times on stream and lower temperatures. This might be an indication of increased carbon deposition and lower catalyst activity when Cu is added. Further, it was reported[7] that Cu is intimately mixed with iron catalysts. During reduction and reaction above 300 °C, it is possible to form Fe−Cu bimetallic clusters, which could reduce catalyst activity. On the basis of another recent study using TPR/temperature-programmed desorption (TPD) and Mossbauer techniques,[2] Cu could suppress iron carbonization, and this might cause lower catalyst activity of the Fe−K/AC catalysts. Finally, Cu was also reported[2] to promote oxidation of iron carbides to Fe_3O_4 and, thus, to accelerate catalyst deactivation. However, this might be not the case for the

Table 1. Activities and HC Selectivities over Fe−K/AC Catalysts with and without Cu[a]

catalyst	15.7Fe−0.94K/AC (0Cu)			15.7Fe−0.8Cu−0.94K/AC (1Cu)				15.7Fe−2Cu−0.94K/AC (2Cu)			
temperature (°C)	260	270	280	260	270	280	290	260	270	280	290
TOS (h)	5−22	22−46	46−70	4−19	19−43	43−67	67−91	5−22	22−46	46−70	70−94
conversion (%)											
X_{CO}	50.7	59.9	85.7	30.7	38.6	45.3	55.5	28.1	38.6	52.5	65.3
X_{H_2}	40.0	44.9	63.0	28.8	33.6	38.2	45.1	23.8	30.6	39.0	46.0
$X_{CO + H_2}$	45.5	52.7	74.8	29.8	36.2	41.9	50.5	26.0	34.8	46.0	56.0
HC selectivity (wt %)											
CH_4	7.8	6.6	8.6	8.9	7.5	8.8	9.4	8.1	7.0	7.9	8.9
C_2-C_4	41.7	34.4	34.9	37.3	31.9	35.6	36.3	37.9	32.5	35.2	36.0
C_{5+}	50.5	59.0	56.5	53.8	60.6	55.6	54.3	54.0	60.5	56.9	55.1
r_{FTS} (mmol/h)	19.40	22.33	31.62	12.51	15.94	21.83	23.74	10.90	15.00	19.85	24.31
r_{WGS} (mmol/h)	16.20	19.77	28.60	8.67	12.08	16.28	21.48	8.81	12.15	17.01	21.53
H_2/CO usage ratio	0.73	0.70	0.68	0.84	0.78	0.76	0.73	0.76	0.71	0.67	0.63
CO_2 selectivity (%)	45.5	47.0	47.5	41.9	43.1	45.2	45.7	44.6	44.9	46.1	47.0
J_p[b]	22.2	27.9	39.8	9.9	7.2	10.5	16.6	10.6	7.5	12.6	21.1
total mass balance (%)		101−98			96−99				101−98		

[a] AC is peat-derived. Metal compositions in wt %. Other process conditions: 3 nL (g of catalyst)$^{-1}$ h^{-1}, 300 psig, and H_2/CO = 0.9. [b] $J_p = (P_{CO_2}P_{H_2})/(P_{CO}P_{H_2O})$.

Table 2. TOF Values of Fe−K/AC Catalysts with and without Cu under the Conditions of Table 1

catalyst	15.7Fe−0.94K/AC (0Cu)			15.7Fe−0.8Cu−0.94K/AC (1Cu)				15.7Fe−2Cu−0.94K/AC (2Cu)			
temperature (°C)	260	270	280	260	270	280	290	260	270	280	290
TOS (h)	5−22	22−46	46−70	4−19	19−43	43−67	67−91	5−22	22−46	46−70	70−94
TOF (s^{-1})	0.15	0.18	0.25	0.09	0.12	0.16	0.19	0.08	0.11	0.15	0.18

carbon-supported Fe catalysts, because the catalyst deactivation rate is no faster on Cu-containing Fe−K/AC catalysts.

The decrease in Fe/AC catalyst activity is consistent with a recent study on precipitated iron catalysts reported by Li and co-workers,[2] but it is not in line with the Cu studies by Bukur et al.,[1] Iglesia and co-workers,[4] and O'Brien and Davis.[5] The discrepancy might arise from the different amounts of Cu used in the catalyst studies, because the reaction conditions used in these studies are similar and represent industrially relevant FTS conditions. In Li and co-workers' study,[2] the Cu level was fixed at about 6 parts per 100 parts of Fe (6 wt % on a Fe basis). However, in the other studies,[1,4,5] the amounts of Cu added were less than 3 wt % on an active-metal basis. In our study, 0.8−2.0 wt % Cu per 15.7 wt % Fe corresponds to a Cu weight fraction of 4.8−12 wt % on an active-metal basis, which is closer to or greater than the amount used by Li and co-workers.[2] Therefore, it is likely that low Cu loadings (<3 wt % on an active-metal basis) increase catalyst activity, while high Cu loadings (>5 wt %) decrease catalyst activity.

As mentioned earlier, the 2Cu catalyst displays higher activity than the 1Cu catalyst at higher temperatures (280 and 290 °C). It may be that, at higher temperatures, more Cu promotes reduction of iron oxides during FTS and leads to more active sites (iron carbides).

Table 1 also shows reduced WGS activities with the addition of Cu on the catalysts. This is quantified by lower values of CO_2 selectivity, partial pressure coefficient, J_p [$\equiv(P_{CO_2}P_{H_2})/(P_{CO}P_{H_2O})$], and H_2/CO usage ratio. The result is in agreement with Li and co-workers.[2] The factors that play a role in the loss of WGS activity are probably the same as those for the loss of FTS activity.

From a kinetic point of view, the effect of the Cu promoter in changing FTS and WGS rates should be reflected by changing activation energy, E_a, and/or the pre-exponential factor, k_0, of the Fe−K/AC catalyst. As reported in the literature,[1,15] the FTS rate can be expressed as first-order in hydrogen partial pressure when syngas conversion is less than 70 wt %

$$r_{FTS} = k_{FTS}P_{H_2} \tag{4a}$$

where k_{FTS} is the apparent FTS reaction rate constant.

The WGS rate can be expressed by[16]

$$r_{WGS} = k_{WGS}P_{CO}P_{H_2O}[1 - P_{CO_2}P_{H_2}/(P_{CO}P_{H_2O}K_{eq})] \tag{4b}$$

where k_{WGS} is the apparent WGS forward rate constant and K_{eq} is the WGS equilibrium constant. The rate constants k_{FTS} and k_{WGS} are temperature-dependent through the Arrhenius relationship

$$k_i = k_{i,0}\exp[-E_{i,a}/(R_gT)] \tag{4c}$$

where i = FTS or WGS. K_{eq} depends upon temperature through[17]

$$K_{eq} = \exp[(4577.8/T) - 4.33] \tag{4d}$$

The apparent pre-exponential factors, $k_{FTS,0}$ and $k_{WGS,0}$, and the apparent activation energies, $E_{FTS,a}$ and $E_{WGS,a}$, are given in Table 3.

From Table 3, the addition of Cu decreases both $E_{FTS,a}$ and $k_{FTS,0}$, as compared to the 0Cu catalysts. The decrease in $E_{FTS,a}$ implies enhancement of FTS by the copper promoter, while the decrease in $k_{FTS,0}$ suggests a (partially) compensating inhibition. When the 1Cu catalyst is compared to the 0Cu catalyst in the temperature range of 260−290 °C, the extent of the FTS rate constant increase resulting from the decreased $E_{FTS,a}$ is a factor of

Table 3. Apparent Kinetic Parameters[a] and Correlation Coefficients R^2 for Fe−K/AC Catalysts with and without Cu

	catalyst[b]	0Cu	1Cu	2Cu
FTS[c]	E_a (kJ/mol)	88.2	66.1	78.2
	k_0 [mmol (g of Fe)$^{-1}$ h^{-1} MPa^{-1}]	6.71×10^{10}	2.74×10^{8}	3.61×10^{9}
	R^2	0.994	0.988	0.997
WGS[d]	E_a (kJ/mol)	189.1	109.1	93.6
	k_0 [mmol (g of Fe)$^{-1}$ h^{-1} MPa^{-2}]	2.74×10^{22}	1.50×10^{14}	5.10×10^{12}
	R^2	0.588	0.733	0.616

[a] $k = k_0 \exp[-E_a/(R_gT)]$. [b] Support is a peat-derived carbon, and all catalysts contain 15.7 wt % Fe and 0.9 wt % K. [c] Assuming $r_{FTS} = k_{FTS}P_{H_2}$ from refs 1 and 10. [d] Assuming $r_{WGS} = k_{WGS}P_{CO}P_{H_2O}[1 - P_{H_2}P_{CO_2}/(P_{CO}P_{H_2O}K_{eq})]$ from ref 16, with $K_{eq} = \exp(4577.8/T - 4.33)$ from ref 17.

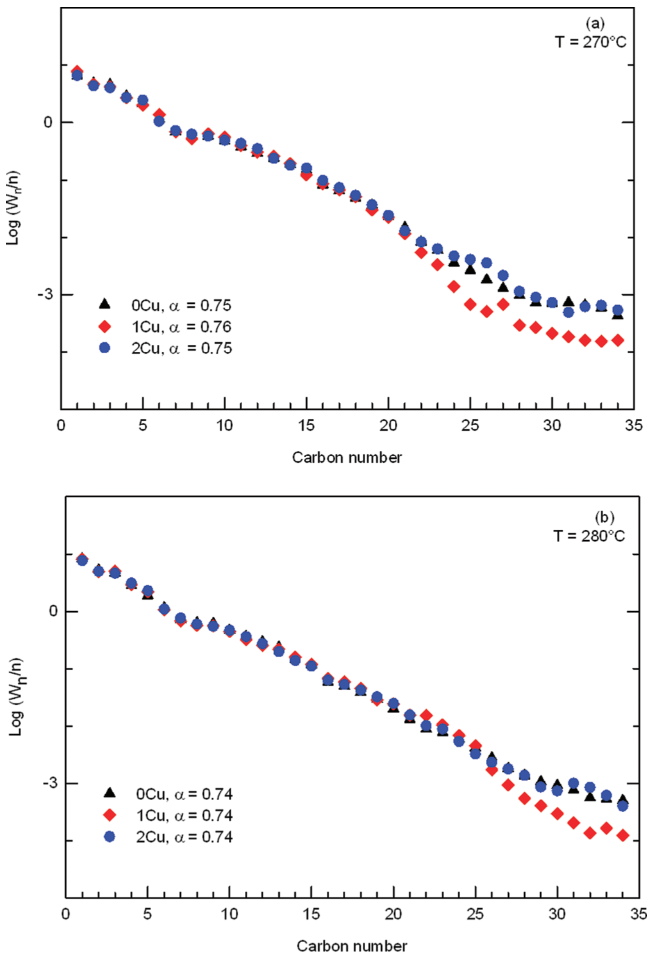

Figure 3. Overall HC distributions over Fe−K/AC catalysts with and without Cu at (a) 270 °C and (b) 280 °C. Other reaction conditions are the same as in Figure 2.

111−146, as compared to a decrease of the rate constant by a factor of 244 resulting from the decreased $k_{FTS,0}$. Similarly, for the addition of 2 wt % Cu, the increase in the rate constant because of the decreased $E_{FTS,a}$ is by a factor of 7.5−8.5, as compared to a decrease from the decreased $k_{FTS,0}$ by a factor of 17.6. Therefore, the Cu addition decreases the FTS catalyst activity resulting from the loss in the pre-exponential factor more than the Cu increases activity because of the loss in activation energy. In contrast, we reported earlier[3] that, while the addition of potassium also decreases both $E_{FTS,a}$ and $k_{FTS,0}$, the effect of decreasing $E_{FTS,a}$ is dominant in that case. It has been noted[2] that a copper promoter increases H_2 adsorption but suppresses CO adsorption, while a potassium promoter increases CO adsorption but decreases H_2 adsorption.

Table 3 shows the effect of Cu on WGS as well. Again, the addition of Cu decreases both $E_{WGS,a}$ and $k_{WGS,a}$. For WGS,

Table 4. Olefin/Paraffin Ratios and Internal Olefin Percentages over Fe−K/AC Catalysts with and without Cu[a]

catalyst[b]	0Cu	1Cu	2Cu	0Cu	1Cu	2Cu
temperature (°C)		260			270	
TOS (h)	21.0	3.0	3.0	85.0	25.0	30.0
$X_{CO + H_2}$ (%)	40.9	34.4	32.0	41.5	41.1	38.5
olefin/paraffin ratio[c]						
C_2	3.2	2.1	1.8	3.9	2.3	3.2
C_4	5.1	3.4	3.2	6.4	3.7	5.2
C_5	2.4	2.3	1.7	3.6	2.3	2.2
internal olefin content[c,d]						
C_4	8.6	10.5	14.3	6.0	7.9	6.2
C_5	11.2	14.3	28.1	7.9	9.5	9.4

[a] Other reaction conditions: 300 psig, H_2/CO = 0.9, and 3 nL (g of catalyst)$^{-1}$ h^{-1}. [b] Support is a peat-derived carbon, and all catalysts contain 15.7 wt % Fe and 0.9 wt % K. [c] From the gas phase. [d] Internal olefin content (%) = 100 × internal olefins/(internal olefin + 1-olefins).

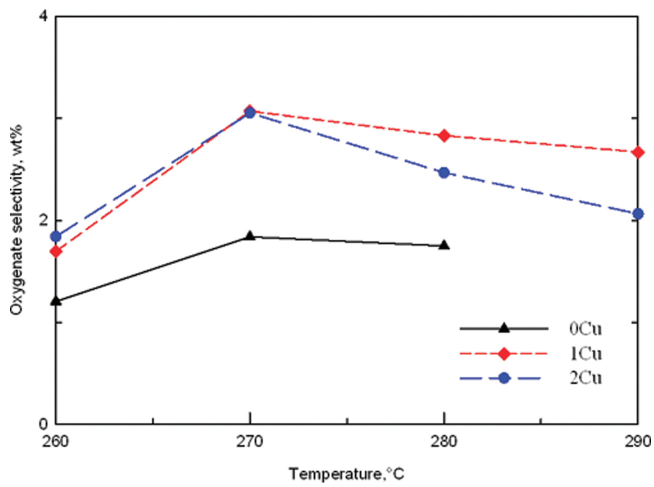

Figure 4. Change of oxygenate selectivity with temperature over Fe−K/AC catalysts with and without Cu. Other reaction conditions are the same as in Figure 2.

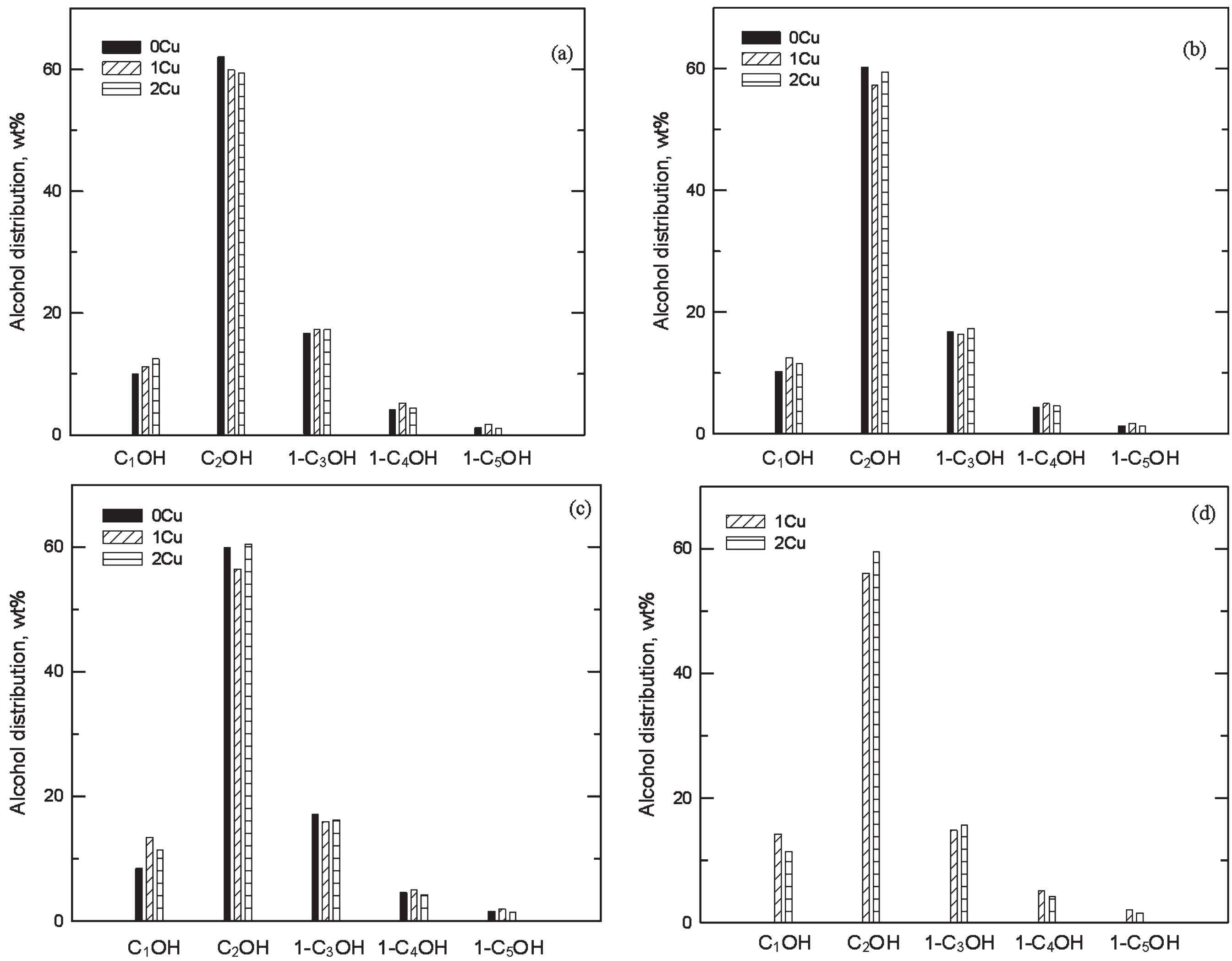

Figure 5. Effect of Cu on alcohol distribution at (a) 260 °C, (b) 270 °C, (c) 280 °C, and (d) 290 °C. Other reaction conditions are the same as in Figure 2.

the addition of Cu decreases the rate because of the decrease in the pre-exponential factor more than the Cu increases the rate because of the loss in activation energy.

3.3. HC Distribution and Selectivities. As before,[3,10,11] HCs up to C_{34} can be easily identified, while C_{35+} chains are not seen. This abrupt cutoff can also be seen in the overall Anderson−Schulz−Flory (ASF) distribution of the Cu-free and Cu-containing Fe−K/AC catalysts shown in Figure 3. The abrupt cutoff is ascribed to the nature of the AC support, i.e., its pore structure and surface morphology. The cutoff is unlikely to be due to heavier HCs being retained in the pores because, as shown earlier as well,[3] our overall mass balances and atomic closers are reasonably good, viz., 98−101%. Rather the physical structure of the AC probably plays an important role in the cutoff of the HC chain length. It has been previously reported[18] that HCs up to C_{20} only were formed over a Co catalyst supported on almond-based AC.

The ASF distribution appears to show that a different effect of Cu addition is seen for the selectivities of very heavy HCs: at 270 °C, for the 1Cu catalyst, amounts of C_{22+} species appear to decrease, whereas for the 2Cu catalyst, the values are similar to those for 0Cu. A similar effect is seen at 280 °C, except that it starts at C_{26+}. However, the amounts of these higher chain HCs are relatively small, and these effects could be subject to GC analytical error.

The effects of copper addition on HC selectivity are quantified in Table 1. The addition of 0.8 wt % Cu to the Fe−K/AC catalyst increases CH_4 selectivity, slightly decreases C_{2-4} selectivity, and increases C_{5+} selectivity. Increasing the Cu loading to 2 wt % does not change these numbers significantly.

Olefin/paraffin ratios, olefin contents, and internal olefin contents over 0Cu, 1Cu, and 2Cu catalysts are compared in Table 4. The comparisons are made under similar conversion levels (32−40 wt % at 260 °C and 39−42 wt % at 270 °C) consistent with the literature.[5,19] The addition of Cu into the Fe−K/AC catalyst decreases C_2, C_4, and C_5 olefin contents and increases the internal olefin content of C_4 and C_5. The change is generally monotonic with Cu loading at 260 °C. At 270 °C, the trend is generally reversed for 2Cu relative to 1Cu. In general, the effect of the Cu promoter is to cause a higher content of secondary olefins. This strongly demonstrates that Cu increases hydrogen adsorption on the catalyst surface and, therefore, increases hydrogenation and/or isomerization, consistent with the literature.[1,2]

3.4. Selectivity to Alcohols. Plots of oxygenate selectivity (defined as the formation rate of oxygenate as a percentage of the total formation rate of oxygenates and HCs) versus the temperature are presented in Figure 4. Although the oxygenate selectivities of the Fe−K/AC catalysts with or without Cu are small, less than 3 wt %, remarkable differences are observed after the addition of Cu. From Figure 4, values of oxygenate selectivity for the 1Cu and 2Cu catalysts are increased by 50−70 wt % compared to that of the 0Cu catalyst. However, the addition of more Cu suppresses the increase. In the case of each catalyst (0Cu, 1Cu, and 2Cu), there is an optimum temperature for maximum oxygenate selectivity; that value appears to be around 270 °C in all cases.

The observed increase in oxygenate selectivity by the Cu promoter is probably due to the Cu promoter attenuating CO adsorption on the surface of the catalyst. HC formation and oxygenate formation are in competition and follow different pathways during FTS. It is generally accepted that the carbide mechanism is responsible for HC formation.[20] Here, the polymerization is initiated by molecular dissociation of CO, and the $-CH_2-$ monomer created by the adsorbed C species (or C*) and the adsorbed H_2 species (or H*) propagates the chain growth. However, for the formation of the oxygenates, the CO insertion mechanism is assumed to be the main path.[21] Here, molecular chemisorption of CO occurs on the metal (M), and hydrogenation forms $M{=}(CO)H_2$, followed by chain growth. As mentioned earlier, the Cu promoter in the 1Cu catalyst probably increases CO molecular adsorption and inhibits CO dissociation. In the 2Cu catalyst, the higher Cu content slightly reduces the increase of oxygenate selectivity. It is possible that the 2Cu catalyst could form more of the Cu−Fe phase during the FTS reaction[7] because of the higher Cu loading (12 wt %). Alternatively, more Cu changes carbonation[2] and/or blocks CO molecular adsorption, resulting in a slight lowering of oxygenate selectivity. The decrease of oxygenate selectivity with the temperature above 270 °C is likely because the higher temperatures negatively affect CO adsorption and increase CO dissociation.

3.5. Distribution of Alcohols. The Cu-containing catalysts also produce alcohols, primarily C_1-C_5 straight chains. Panels a−d of Figure 5 show the changes of the alcohol distributions with Cu loadings at 260−290 °C. Clearly, ethanol is the dominant oxygenate in all cases (approximately 60%), followed by C_3-OH (15%), C_1-OH (10%), C_4-OH, and C_5-OH. Again, this is consistent with the results reported previously over Fe−K/AC and Mo−Ni−K/AC catalysts.[3,22]

Overall, Cu addition does not significantly change alcohol distribution. However, a slight change in alcohol distribution by the Cu promoter can be seen at different temperatures. At 260 °C, the addition of 0.8 wt % Cu increases methanol, propanol, and higher alcohols and decreases ethanol. As temperatures are raised to 270 °C, Cu addition still increases the contents of methanol, C_4-OH, and C_5-OH but the trend of increasing C_3-OH and decreasing C_2-OH is weakened. When the temperatures are raised above 280 °C, an increase in C_2-OH and a decrease in C_3-OH are found. In all cases, increasing Cu loading from 0.8 to 2.0 wt % after 270 °C results in slight increases in C_2-OH and C_3-OH but decreases in C_4-OH and C_5-OH .

The relatively small changes in alcohol distribution with Cu content and temperature might be due to some iron carbonization occurring at different temperatures during FTS, as pointed out previously.[3]

4. CONCLUSION

The addition of 0.8−2.0 wt % copper significantly promotes the reduction of 15.7 wt % Fe−0.9 wt % K/AC catalysts via decreased reduction temperatures and increased reduction rates. Copper also affects catalyst behavior by changing the catalyst activity and selectivities to HCs (olefin/paraffin and 1-olefins/2-olefins) and oxygenates. These have been explained in terms of the altering of the surface chemistry of the Fe−K/AC catalysts by Cu. The Cu added in the present work corresponds to weight fractions of 5−12 wt % on an active-metal basis.

Copper added to the Fe−K/AC catalyst decreases both FTS and WGS activities. The lower Cu loading may lead to the formation of some Fe−Cu clusters, and/or the Cu promoter could suppress carbonization of Fe[2] and/or foster carbon deposition during the carbonization at the beginning of FTS.[7] Cu decreases both the activation energy and pre-exponential factor for FTS, but it predominately affects the latter parameter. For WGS, Cu affects the activation energy and pre-exponential factor similarly. Increasing the Cu loading from 0.8 to 2.0 wt % results in a decreased change in the activity of the Fe−K/AC catalyst.

The addition of Cu does not significantly change the overall HC distribution of the Fe−K/AC catalyst, but a change in the amounts of the longer chain products may be noted. An increase in the internal olefin content and a reduction in the olefin content are observed under similar levels of CO conversion. This confirms that Cu promotes hydrogen adsorption and enhances hydrogenation and isomerization reactions of iron-based catalysts.

Copper increases the oxygenate selectivity of the Fe−K/AC catalysts during FTS. This result indicates that the Cu promoter has the role of increasing molecular CO adsorption and decreasing CO dissociation on the surface of the catalyst. Cu slightly changes the distribution of alcohols, tending to increase methanol and C_3-C_5 alcohols but decreasing ethanol content.

■ AUTHOR INFORMATION

Corresponding Author

*Telephone: (304) 293-9337. Fax: (304) 293-4139. E-mail: dady.dadyburjor@mail.wvu.edu.

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